

# 08 — Mu-Calculus

Fixpoints, Modalities, and Common Patterns

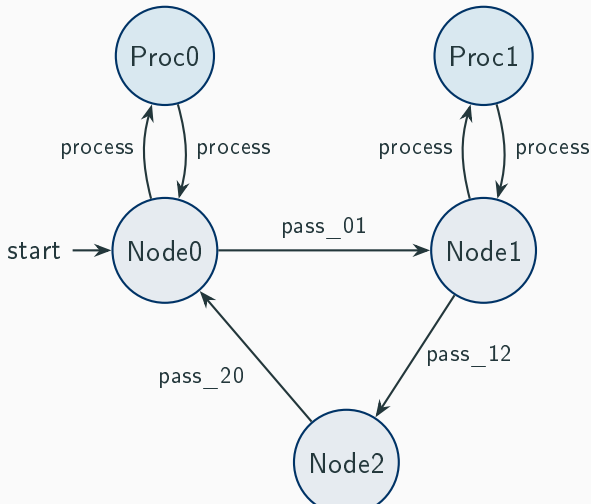
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Mununu Tutorial

# The Token Ring

A token circulates among three nodes; each can **process** locally before passing it on.



# Operator Reference

| Syntax                                        | Name              | Meaning                                   |
|-----------------------------------------------|-------------------|-------------------------------------------|
| $\text{nu } X. \varphi$                       | Greatest fixpoint | Safety / invariance                       |
| $\text{mu } X. \varphi$                       | Least fixpoint    | Reachability / liveness                   |
| $[] \varphi$                                  | Box               | <i>All</i> successors satisfy $\varphi$   |
| $\langle \rangle \varphi$                     | Diamond           | <i>Some</i> successor satisfies $\varphi$ |
| $[ \text{labels}=\{a\} ] \varphi$             | Labeled box       | All $a$ -successors satisfy $\varphi$     |
| $\langle \text{labels}=\{a\} \rangle \varphi$ | Labeled diamond   | Some $a$ -successor satisfies $\varphi$   |
| $\varphi \ \&\& \ \psi$                       | Conjunction       | Both hold                                 |
| $\varphi \    \ \psi$                         | Disjunction       | At least one holds                        |
| $! \ \varphi$                                 | Negation          | Does not hold                             |
| $\text{true} / \text{false}$                  | Constants         | Trivially satisfied / unsatisfied         |

# Safety and Reachability

```
1 // == Greatest fixpoint (nu) --Safety/Invariance ==
2
3 // nu X. ([] X) --"all reachable states are valid"
4 // The canonical safety invariant.
5 formula safety_invariant {
6     over TokenRing;
7     body = nu X. ([] X);
8 }
9
10 // == Least fixpoint (mu) --Reachability ==
11
12 // mu X. (Node2 || <> X) --"Node2 is reachable from here"
13 // Computes the set of states from which Node2 can be reached.
14 formula node2_reachable {
15     over TokenRing;
16     body = mu X. (Node2 || <> X);
17 }
```

# Labeled Modalities

```
1 // == Labeled box modality ==
2
3 // [ labels = { pass_01 } ] Node1
4 // "Every pass_01 transition leads to Node1"
5 // True for all states (pass_01 only exists from Node0 -> Node1).
6 formula pass01_leads_to_node1 {
7     over TokenRing;
8     body = [ labels = { pass_01 } ] Node1;
9 }
10
11 // == Labeled diamond modality ==
12
13 // < labels = { pass_12 } > Node2
14 // "There exists a pass_12 transition leading to Node2"
15 // True only for Node1 (the only state with a pass_12 transition).
16 formula can_pass_to_node2 {
17     over TokenRing;
18     body = < labels = { pass_12 } > Node2;
```

## Nested Fixpoints — GF Pattern

```
1      // === Nested fixpoints --GF pattern ===
2
3      // nu X. (mu Y. (Node0 || <> Y)) && ([] X)
4      // "Infinitely often Node0" (GF Node0)
5      // From every reachable state, Node0 can always be reached again.
6      formula gf_node0 {
7          over TokenRing;
8          body = nu X. ((mu Y. (Node0 || <> Y)) && ([] X));
9      }
```

$\nu X. (\mu Y. (\text{Node0} \vee \Diamond Y)) \wedge [] X$

- **Outer  $\nu$**  (safety): the property holds at every reachable state.
- **Inner  $\mu$**  (reachability): Node0 can be reached from the current state.
- **Combined**: “from every reachable state, Node0 can always be reached again” — the **GF (infinitely often)** pattern.

# Common Patterns Summary

| Pattern                  | Formula                                                    |
|--------------------------|------------------------------------------------------------|
| Safety invariant         | $\nu X. []X$                                               |
| Reachability             | $\mu X. (p \vee \Diamond X)$                               |
| Labeled transition check | $[labels = \{a\}] q$ or $\langle labels = \{a\} \rangle q$ |
| GF (infinitely often)    | $\nu X. (\mu Y. (p \vee \Diamond Y)) \wedge []X$           |
| No deadlock              | $\nu X. \Diamond \text{true} \wedge []X$                   |

## Key insight:

- $\nu$  (nu) starts from “everything satisfies” and removes violations — **safety**.
- $\mu$  (mu) starts from “nothing satisfies” and adds witnesses — **reachability**.
- Nesting them yields **liveness** properties (e.g., infinitely often).